

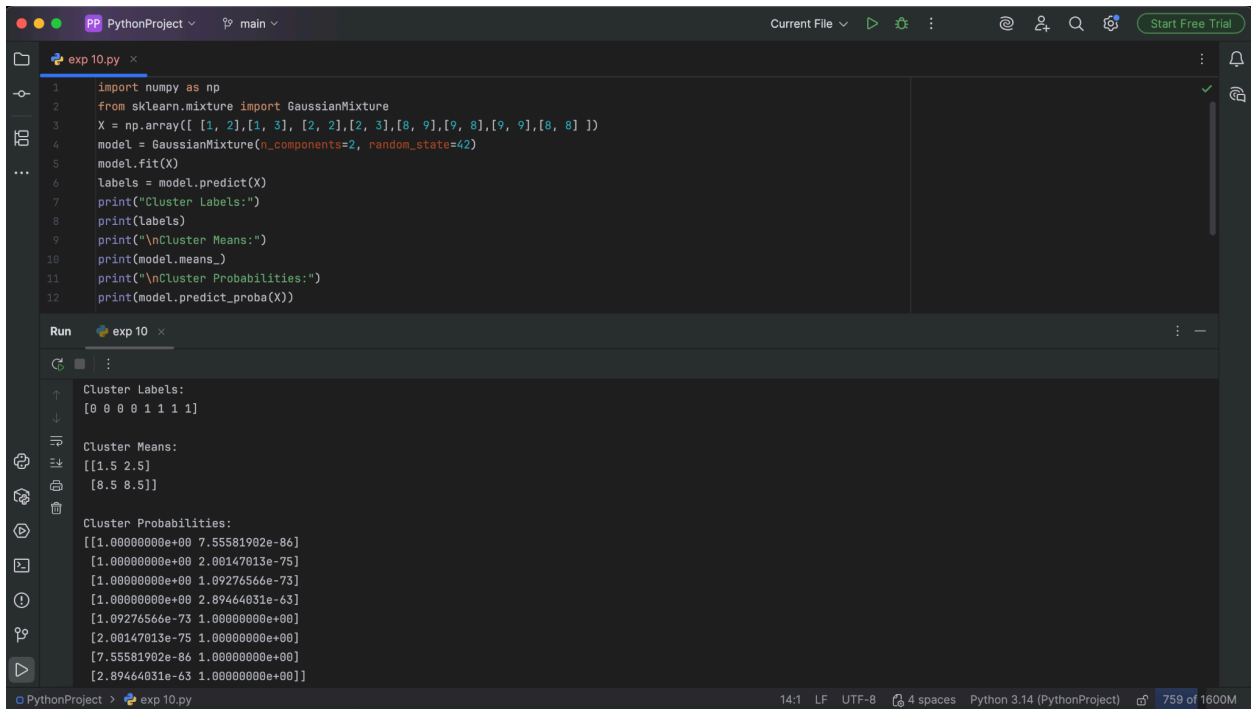
## EXPERIMENT - 10

**Aim:** To implement the Expectation-Maximization algorithm in Python for finding parameters of data with hidden variables.

### Procedure:

1. Import the required libraries.
2. Load or generate the dataset.
3. Initialize the model parameters.
4. Perform the Expectation (E) step to estimate the probability of each data point belonging to each cluster.
5. Perform the Maximization (M) step to update the model parameters.
6. Repeat the E-step and M-step.
7. Continue until the parameters converge.
8. Display the final clusters or estimated parameters.

### Output:



```
1 import numpy as np
2 from sklearn.mixture import GaussianMixture
3 X = np.array([ [1, 2], [1, 3], [2, 2], [2, 3], [8, 9], [9, 8], [9, 9], [8, 8] ])
4 model = GaussianMixture(n_components=2, random_state=42)
5 model.fit(X)
6 labels = model.predict(X)
7 print("Cluster Labels:")
8 print(labels)
9 print("\nCluster Means:")
10 print(model.means_)
11 print("\nCluster Probabilities:")
12 print(model.predict_proba(X))
```

Run exp 10

```
Cluster Labels:
[0 0 0 0 1 1 1 1]

Cluster Means:
[[1.5 2.5]
 [8.5 8.5]]

Cluster Probabilities:
[[1.00000000e+00 7.55581902e-86]
 [1.00000000e+00 2.00147013e-75]
 [1.00000000e+00 1.09276566e-73]
 [1.00000000e+00 2.89464031e-63]
 [1.09276566e-73 1.00000000e+00]
 [2.00147013e-75 1.00000000e+00]
 [7.55581902e-86 1.00000000e+00]
 [2.89464031e-63 1.00000000e+00]]
```

**Result:** The Expectation-Maximization algorithm successfully estimated the model parameters and grouped the data into clusters.